

ceramic dual in-line package (SBDIP), and can stand operating temperatures of -40°C to $+175^{\circ}\text{C}$, with a guaranteed 1000 hours life.

Instrumentation amplifiers. Companies like TI and ADI have high-temperature instrumentation amplifiers for various applications. ADI's AD8229 is an ultra-low noise instrumentation amplifier designed for measuring small signals in the presence of large common-mode voltages and at high temperatures in the range of 210°C . The process is dielectrically isolated to avoid leakage currents at high temperatures, and the design architecture compensates for the low base-emitter voltages at high temperatures.

Capable of measuring even tiny signals, the AD8229 is available in an eight-lead SBDIP. For space-constrained applications, there is also an eight-lead plastic standard small outline package (SOIC). It is used extensively for down-hole instrumentation, harsh-environment data acquisition, exhaust gas measurements and vibration analysis.

TI also has a range of high-temperature rated instrumentation amplifiers, such as the INA128-HT, INA129-HT and INA333-HT, which can operate from -55°C to $+210^{\circ}\text{C}$. TI's high-temperature products utilise highly-optimised silicon (die) solutions with design and process enhancements to maximise performance over extended temperatures. The INA129 is available in an eight-pin ceramic dual inline package (DIP) and an eight-pin ceramic surface-mount package, specified for the -55°C to 210°C temperature range. The INA128 is available in an eight-pin small outline surface-mount package (SO8), specified for the -55°C to $+175^{\circ}\text{C}$ temperature range.

Flash memory. TI has introduced what it claims to be the industry's first high-temperature 4MB Flash memory device. TI's SM28VLT32-HT has an operating temperature ranging from -55°C up to $+210^{\circ}\text{C}$, designed for data logging and firmware designs. It comes in a small ruggedised packaging—14-pin ceramic flat package and KGD.



SM28VLT32-HT 4MB Flash memory device by TI (Courtesy: <http://newscenter.jp.ti.com>)

Consumer products are also now featuring temperature-tolerant memories, although the range is much lower than for critical applications. SanDisk, for example, is now promoting its Extreme SDXC UHS-I card and Extreme Pro MicroSDHC UHS-I 16GB card for use in cameras, video recorders, etc, and the Cruiser Glide USB Flash drive, which can be used in cold and hot weather spanning -25°C to $+85^{\circ}\text{C}$.

Applications processor. One example of a high-temperature applications processor is TI's OMAPL137-HT. It is a 300MHz dual-core (ARM926EJ-S and TMS320C674x DSP) applications processor designed for use from -55°C to $+175^{\circ}\text{C}$, enabling harsh environment intelligent electronic processing. It runs a high-level, real-time operating system, and has robust 1.8V/3.3V peripherals (Ethernet MAC, EMIF 1/2, USB2.0 OTG, McASP). It comes in a 176-pin plastic quad-flat-pack package.

Auto-grade products. Companies like National Semiconductor, NXP, Altera and Bosch offer 'automotive-grade products,' which generally means that these adhere to Automotive Electronics Council (AEC) standards for humidity, temperature, stress and other factors. There are various levels of auto compliance such as Q100, Q200 and so on. NXP, for example, complies with AEC-Q100. Within AEC-Q100, there are five grades (0-5) for operating temperature, representing lower limits from -40°C to $+150^{\circ}\text{C}$ on the upper end. So when choosing an auto-grade product, one needs to check its compliance with the required standard/grading.

Test and measurement systems. When using electronics for critical applications, reliability is very important. In order to test and ensure non-failure,

constant test and measurement is required, which means that the test system for such electronics should also be able to endure the same temperature conditions as the electronic components. NI's CompactRIO is designed for such applications in harsh environments and small places.

"Size, weight and I/O channel density are critical design requirements in many such embedded applications. Taking advantage of the performance and small size of FPGA devices, CompactRIO is able to deliver unprecedented control and acquisition capabilities in a compact, rugged package with extreme industrial certifications and ratings for operation in harsh industrial environments," says Raghavan.

With a variety of connectivity options, NI CompactRIO can be interfaced with most of the industrial sensors like pressure transducers, magnetic pickup sensors, digital encoders, temperature sensors, accelerometers, nuclear densitometers and magnetic flow meters. It is programmed with NI LabVIEW graphical programming tools. A bare-board version of the CompactRIO, the NI Single Board RIO, is also available.

"Korea Gas Corporation has used CompactRIO to develop a safety management system to monitor exposed city gas pipelines attached to bridges. Ventura Aerospace was able to develop an intelligent fire monitoring and suppression control system for FedEx Express using NI LabVIEW software and NI Single-Board RIO hardware to prevent catastrophic fires within freight aircraft and keep pilots, packages and planes safe from fires that may start in shipping containers. Similarly, Lime Instruments, LLC, built